

STROMMINGSBADN, Finland

WMO: 027810

Lat: 62.979N

Lon: 20.740E

Elev: 12

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.0	7.3	-4.3	4.4	-0.1	3.5	6.6	7.1	39.9	31.0	37.3	29.8	11.2	50	0.862

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	71.3	67.5	68.0	64.3	65.6	62.3	67.7	71.2	65.0	67.6	63.1	65.2	10.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	95.8	70.1	63.9	88.9	67.2	62.0	83.2	64.8	32.1	71.7	30.1	67.4	28.6	65.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	33.4	30.2	0.0	73.4	10.4	3.9	-7.5	76.3	-13.6	78.5	-19.4	80.7	-27.0	83.6		
(5)			0.9	67.7	11.1	3.1	-7.1	69.9	-13.7	71.8	-19.9	73.5	-28.0	75.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	26.3	25.5	29.6	35.6	44.8	52.8	60.2	60.6	54.6	44.5	37.7	32.1
	DBStd	13.60	8.04	9.89	5.85	3.46	5.02	4.21	4.78	4.06	4.44	4.58	6.00	7.89
	HDD50	3798	736	686	632	432	179	18	0	0	14	177	368	556
	HDD65	8381	1201	1106	1097	882	627	367	163	149	313	637	818	1021
	CDD50	920	0	0	0	0	17	102	317	328	151	5	0	0
	CDD65	26	0	0	0	0	0	0	15	11	0	0	0	0
	CDH74	24	0	0	0	0	0	0	14	10	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.7	17.5	16.7	15.4	13.4	13.6	13.3	12.6	13.9	15.9	17.9	18.6	19.3
Precipitation	PrecAvg	29.7	2.6	2.0	1.9	1.6	1.9	2.4	3.1	3.1	2.6	2.8	2.9	2.8
	PrecMax	38.3	4.5	3.8	3.5	3.1	4.3	4.0	5.9	5.3	5.6	6.4	5.6	4.9
	PrecMin	23.3	0.4	0.2	0.5	0.5	0.4	0.9	0.6	1.0	1.0	0.4	0.9	0.7
	PrecStd	3.9	1.0	0.9	0.7	0.7	0.9	0.8	1.3	1.3	1.1	1.3	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.6	38.6	39.5	47.5	60.9	66.4	76.1	74.7	64.4	55.5	49.8	44.6
		MCWB	37.8	36.1	36.6	43.4	55.7	61.4	69.7	69.4	61.8	52.4	47.4	42.8
	2%	DB	38.2	37.2	38.0	44.0	57.2	63.1	72.2	71.4	61.9	53.7	48.2	42.9
		MCWB	36.5	35.3	35.7	41.4	52.7	59.7	68.2	67.4	59.0	51.1	46.7	41.0
	5%	DB	36.9	35.8	37.0	41.6	54.1	60.4	69.2	67.7	60.9	51.4	46.5	41.4
		MCWB	35.2	34.4	35.0	39.3	50.5	57.2	66.0	63.3	58.2	49.1	45.5	39.2
	10%	DB	35.2	34.8	36.0	39.9	51.6	58.3	66.6	65.5	60.0	50.1	44.7	40.0
		MCWB	33.5	33.7	34.4	37.7	48.4	55.4	63.5	61.8	57.5	48.0	43.2	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.6	36.6	37.1	43.6	56.3	62.0	70.8	71.0	61.4	53.5	47.9	43.4
		MCDB	39.2	37.7	38.5	45.4	60.4	65.6	73.7	74.0	63.4	54.6	48.9	44.2
	2%	WB	36.7	35.3	36.1	40.6	52.8	60.1	68.8	66.8	60.1	51.7	46.8	41.4
		MCDB	37.6	36.3	37.2	42.6	56.4	62.7	72.0	70.7	61.5	53.3	47.9	42.9
	5%	WB	35.3	34.4	35.4	39.0	50.8	58.1	66.4	64.4	59.1	49.7	45.7	39.6
		MCDB	36.8	35.3	36.5	40.5	53.9	60.0	69.4	67.0	60.5	51.1	46.7	41.0
	10%	WB	34.0	33.7	34.6	37.6	48.7	56.2	64.3	62.6	57.8	48.5	43.8	38.3
		MCDB	35.0	34.4	35.8	38.9	51.3	58.1	66.3	64.6	59.6	50.0	44.7	39.7
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.2	5.7	4.7	5.6	5.9	5.5	4.7	4.1	3.9	4.4	5.0
		MCDBR	5.3	4.1	4.6	6.4	8.9	8.7	8.1	5.9	4.8	3.3	4.0	4.6
		MCWBR	4.8	3.6	3.6	4.3	5.8	5.6	5.2	4.4	3.7	3.9	3.9	4.7
	5% WB	MCDBR	4.8	3.8	4.3	5.8	8.9	7.7	7.5	5.5	4.4	3.2	3.8	4.6
		MCWBR	4.5	3.8	3.4	4.2	6.1	5.4	5.2	4.7	3.8	3.9	3.6	4.7
Clear-Sky Solar Irradiance	taub	0.243	0.297	0.317	0.356	0.355	0.352	0.374	0.362	0.336	0.319	0.384	0.349	
	taud	2.075	2.192	2.265	2.319	2.447	2.498	2.462	2.484	2.536	2.391	2.678	2.554	
	Ebn at Noon	123	201	247	261	273	277	266	259	245	196	114	69	
	Edh at Noon	10	20	27	32	31	30	30	27	22	17	9	6	
All-Sky Solar Radiation	RadAvg	46	225	704	1214	1690	1896	1729	1322	778	298	72	24	
	RadStd	6	44	83	124	157	134	148	113	72	57	14	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page